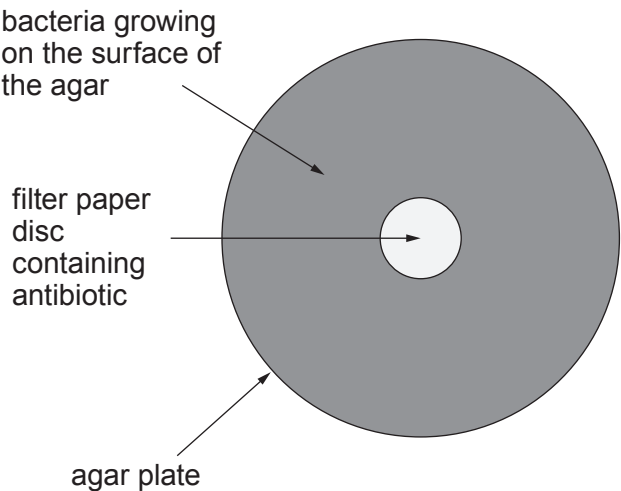


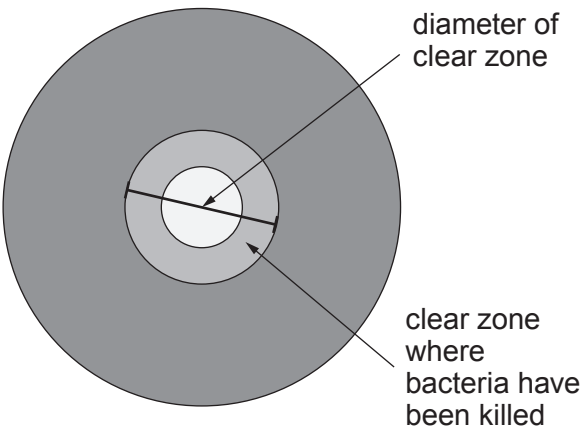
6. Mair investigated four antibiotics (**A**, **B**, **C** and **D**) to compare how effective they were at killing bacteria.
- She set up four agar plates with bacteria growing on them.
 - She placed a filter paper disc containing a different antibiotic in the centre of each plate.
 - She put the plates in an incubator.
 - After three days, she measured the diameter of the clear zone where bacteria had been killed.
 - She calculated the area of the clear zone.

Image 6.1

Agar plate at start of the investigation



Agar plate at three days



- (a) Mair used aseptic techniques to set up the agar plates.
- (i) Explain why it was necessary for her to use **sterilised** forceps to place the filter paper discs on the agar. [1]
-
-
- (ii) Describe how she would have made sure that no bacteria from the environment could enter the agar plates after the filter paper discs were placed on the agar. [1]
-
-
- (b) **Circle** the most suitable temperature for the incubator in a **school** laboratory from the choice below. [1]

10 °C 25 °C 37 °C 60 °C



(c) **Table 6.2** shows Mair’s results.

Table 6.2

	Antibiotic A	Antibiotic B	Antibiotic C	Antibiotic D
Area of clear zone on agar plate (mm ²)	270	360		450

(i) The **radius** of the clear zone for antibiotic **C** was 11 mm.

Complete Table 6.2 by calculating the area of the clear zone for antibiotic **C**, using the formula below, where r = radius.

Area of circle = πr^2 ($\pi = 3.14$)

Give your answer to the **nearest whole number**.
Space for working

[3]

(ii) Use the data in **Table 6.2** to complete **Bar chart 6.3** by:

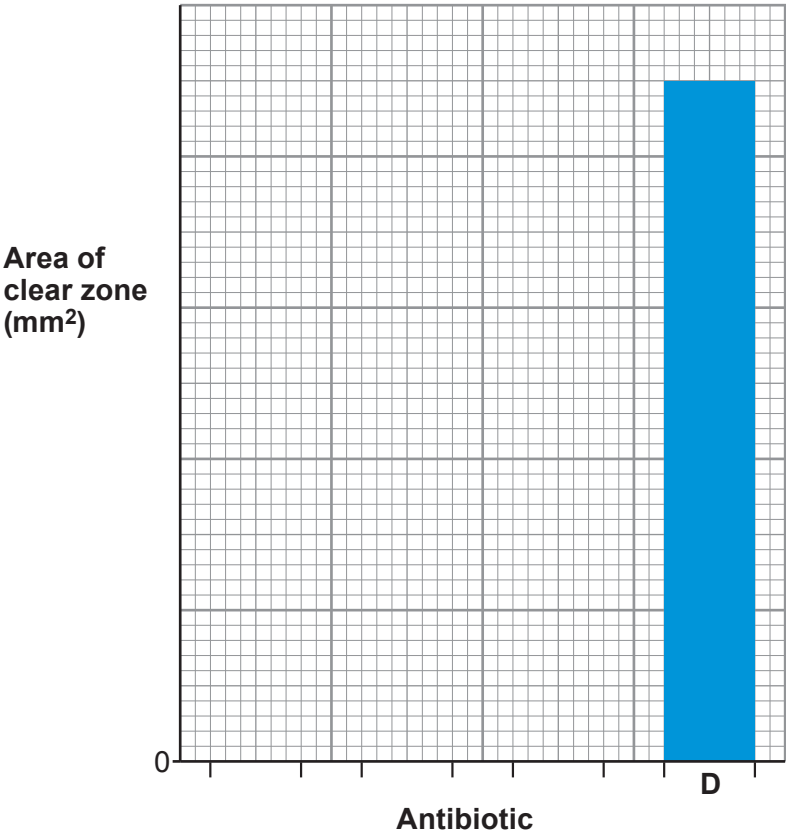
[3]

- I. adding the scale for the area of the clear zone .
- II. drawing bars for antibiotics **A**, **B** and **C** and labelling your bars.



Examiner
only

Bar chart 6.3



(iii) From Bar chart 6.3, state which **one** of the antibiotics was the most effective and explain the reason for your choice. [3]

Antibiotic

Reason

.....
.....

(d) (i) State how Mair could check that her results were reproducible. [1]

.....
.....

(ii) Suggest **one** way in which the investigation could be extended to give more information about the effects of antibiotics on the growth of bacteria. [1]

.....
.....

